SUPPLEMENT: THE ZARISKI TOPOLOGY

Throughout, let k be an algebraically closed field. If a variety is defined over a smaller field K, the definitions below are applied to its geometric points over $\overline{K}$. The purpose of this sheet is only to explain the topological language used in the lectures: closed and open subsets, dense open subsets, and the special simplifications that occur for algebraic curves.

1. Closed and open subsets

Definition 1 (Affine Zariski topology)**.** For a set $S \subseteq k[X_1, \ldots, X_n]$, put

$$V(S) = \{P \in \mathbb{A}^n(k) : f(P) = 0 \text{ for every } f \in S\}.$$

The *Zariski closed subsets* of $\mathbb{A}^n(k)$ are the algebraic sets $V(S)$. Their complements are called *Zariski open subsets.*

These sets form a topology because

$$V(0) = \mathbb{A}^n, \qquad V(1) = \varnothing,$$
$$\bigcap_\alpha V(I_\alpha) = V\left(\sum_\alpha I_\alpha\right), \qquad V(I) \cup V(J) = V(IJ).$$

Thus arbitrary intersections and finite unions of closed sets are closed.

If $X \subseteq \mathbb{A}^n$ is an affine algebraic set, the Zariski topology on X is the *subspace topology*: a subset $Z \subseteq X$ is closed precisely when

$$Z = X \cap V(I)$$

for some ideal $I \subseteq k[X_1, \ldots, X_n]$. We often write this closed set as $V_X(I)$.

Definition 2 (Basic affine open sets)**.** If $f \in k[X]$, define

$$D_X(f) = \{P \in X : f(P) \neq 0\} = X \setminus V_X(f).$$

The sets $D_X(f)$ form a basis for the Zariski topology: every open subset of X is a union of sets of this form.

Example 3. In $\mathbb{A}^2$, the set $V(XY)$ is the union of the two coordinate axes. Its complement is

$$D(XY) = D(X) \cap D(Y),$$

the set of points for which both coordinates are nonzero. The set $D(X)$ is the complement of the vertical axis $X = 0$.

Definition 4 (Projective Zariski topology)**.** A subset of $\mathbb{P}^n(k)$ is Zariski closed if it is the common zero locus of a collection of homogeneous polynomials. If $X \subseteq \mathbb{P}^n$ is a projective algebraic set, its closed subsets are the intersections of X with projective algebraic sets.

For a homogeneous polynomial F, define the basic projective open set

$$D_{+,X}(F) = \{P \in X : F(P) \neq 0\}.$$

The condition $F(P) \neq 0$ is independent of the choice of homogeneous coordinates.

The usual affine charts are basic open sets:

$$U_i = \{[x_0 : \cdots : x_n] : x_i \neq 0\} = D_+(x_i) \simeq \mathbb{A}^n.$$

The Zariski topology on U_i under this identification is exactly its usual affine Zariski topology. Thus projective varieties are covered by affine open subsets, and topological questions can usually be checked chart by chart.

2. Closure, density, and irreducibility

Definition 5. Let $S \subseteq X$. Its *Zariski closure* $\overline{S}$ is the smallest closed subset of X containing S. The set S is *dense* in X if $\overline{S} = X$.

For an affine algebraic set X and a subset $S \subseteq X$, let

$$I_X(S) = \{f \in k[X] : f(P) = 0 \text{ for every } P \in S\}.$$

Then

$$\overline{S} = V_X(I_X(S)).$$

In particular, S is dense in X precisely when every regular function on X which vanishes on S vanishes on all of X.

Definition 6. A nonempty topological space X is *irreducible* if it cannot be written as the union of two proper closed subsets.

For an algebraic set, this agrees with the earlier algebraic definition: X is irreducible if and only if $I(X)$ is prime. Hence a variety is irreducible by definition.

Proposition 7. *For a nonempty topological space X, the following are equivalent:*

(1) X is irreducible;
(2) every nonempty open subset of X is dense;
(3) any two nonempty open subsets of X have nonempty intersection.

This explains several phrases used in the discussion of rational maps. If V is a variety, every nonempty open subset of V is automatically dense. Moreover, two rational expressions which agree on a nonempty open subset agree wherever both are defined, because their difference is a rational function vanishing on a dense set.

Remark 8. The Zariski topology is very different from the Euclidean topology. It is usually not Hausdorff. For example, on an irreducible variety any two nonempty open sets meet, so two distinct points cannot have disjoint open neighbourhoods.

3. The special case of curves

Proposition 9. *Let C be an irreducible algebraic curve over k. Every proper Zariski closed subset of C is finite. Consequently, the nonempty open subsets of C are exactly the complements of finite sets.*

Indeed, a proper closed subset of the one-dimensional irreducible variety C has dimension zero, and a zero-dimensional algebraic set over an algebraically closed field consists of finitely many points.

Thus the Zariski topology on an irreducible curve is the *cofinite topology*:

$$U \subseteq C \text{ is nonempty and open} \quad \Longleftrightarrow \quad C \setminus U \text{ is finite.}$$

This is the main simplification relevant for elliptic curves.

Example 10. The closed subsets of $\mathbb{A}^1(k)$ are $\mathbb{A}^1(k)$ itself and finite sets of points. Indeed, every ideal of $k[X]$ is principal, and a nonzero polynomial has only finitely many roots. The same conclusion holds for $\mathbb{P}^1(k)$.

Example 11. Let E be a smooth projective plane cubic. Then E is an irreducible projective curve. Hence

$$E \setminus \{P_1, \dots, P_r\}$$

is open and dense for every finite set of points. In particular, the affine part of a Weierstrass cubic is

$$E \setminus \{O\},$$

where O is the point at infinity, and this affine part is a dense open subset of E.

A rational function on a curve is regular away from finitely many points. Similarly, a rational map from a smooth projective curve is initially defined on the complement of finitely many points. In many situations relevant to elliptic curves, one then proves separately that the map extends across those missing points.

4. Continuity of algebraic maps

Proposition 12. *A morphism of algebraic varieties is continuous for the Zariski topology.*

For example, if $\phi : X \to Y \subseteq \mathbb{A}^m$ is given by regular functions $\phi_1, \dots, \phi_m$, then

$$\phi^{-1}(V_Y(F_1, \dots, F_r)) = V_X(F_1(\phi_1, \dots, \phi_m), \dots, F_r(\phi_1, \dots, \phi_m)),$$

which is closed. The projective statement follows by working on affine charts.

Remark 13. For the lectures on elliptic curves, the facts to remember are:

- closed subsets are zero loci of polynomial equations;
- basic open sets are obtained by requiring one polynomial not to vanish;
- every nonempty open subset of a variety is dense;
- on an irreducible curve, removing finitely many points gives an open dense subset;
- morphisms are continuous, while rational maps are represented by algebraic formulas on dense open subsets.